

Climate Volatility in Daily Operations

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Panel Discussion

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Why This Conversation, and Why Now

- Efficiency and carbon accounting are mature disciplines. Assessing operational risk from environmental stress is not — there is no shared method for relating exposure to consequence.
- Heat, water constraint, wildfire smoke, and grid instability have become daily operating variables rather than long-range planning inputs.
- The largest build cycle in the industry's history is being designed against local codes and historical climate baselines that increasingly understate 20–30 year operating conditions.
- The same stressor produces very different outcomes at two facilities. Cooling architecture, control tuning, commissioning state, maintenance condition, and staffing depth decide which.
- Operators, hyperscalers, and vendors each see one slice of this problem, and rarely compare notes in the same room. That is what this session is for.

The Gap in Current Frameworks

- ASHRAE TC 9.9 defines IT thermal envelopes — not how external stress consumes the margin to them.
- EN 50600 / ISO-IEC 22237 define facility design and availability class — not climate-driven drift of the design basis over asset life.
- ISO 22301 covers continuity management — not environmental hazard characterization.
- PUE and WUE (ISO-IEC 30134) track efficiency — a facility can hold both metrics while operating with no redundancy left.
- Uptime Tier classifies topology — environmental stress is a common-mode threat that can defeat topological redundancy simultaneously.
- IFRS S2 requires physical climate risk disclosure — and assumes an asset-level assessment method that does not yet exist.
- None of them provides an open method linking environmental exposure to operational consequence. That is the gap.

A Layered Framework

L1 — **External Hazard**
Heat, drought, smoke, grid disturbance — the stressor itself

L2 — **Site & Campus Exposure**
Microclimate, watershed, airshed, transmission region

L3 — **Gray-Space Response**
Chillers, towers, filtration, UPS, generation, controls

L4 — **White-Space Environment**
Rack inlet, coolant supply, airflow, contamination

L5 — **Operational Outcome**
Derating, margin loss, maintenance burden, continuity risk

L6 — **Orchestration (conditional)**
Workload migration — acts backward on L3 and L4

Stress enters at L1 and propagates downward, consuming resilience margin at each layer. L6 is present only where an operator runs distributed capacity — and it is the only layer that acts backward.

Where the Margin Actually Goes

- L1 Hazard — five consecutive days at 41 °C peak dry-bulb with coincident wet-bulb of 24 °C, above the facility's design basis.
- L2 Site exposure — the site runs roughly 1.5 °C above the reference station because of urban heat island effect.
- L3 Gray space — cooling tower approach widens, condenser water rises above design, available chiller capacity falls.
- L4 White space — rack inlet temperature rises into the upper allowable class, but stays within limits.
- L5 Outcome — no service interruption. Availability reporting records a normal period. The redundancy margin is gone.
- L6 Orchestration — latency-tolerant workload migrates out, restoring physical margin at L3 — if the destination is not stressed by the same event.
- The significant outcome is not a failure. It is the silent elimination of margin, which appears in no availability or efficiency metric.

Discussion Themes

- **What has actually changed**
  - Which environmental conditions moved from planning assumption to daily operating variable — and what did you change in response?
- **Where design meets operations**
  - What do operators know about behaviour under stress that never reaches the design basis or the vendor specification?
- **The vendor question**
  - What conditions should equipment be qualified against, and who decides? Where does off-design product behaviour set facility-level margin?
- **Evidence without disclosure**
  - Can open environmental data give the industry a comparable exposure baseline without anyone sharing proprietary telemetry?
- **What should exist that doesn't**
  - What is the single most useful thing OCP could publish next on operational resilience?

What You'll Take Away

- A shared vocabulary — stressor, exposure, vulnerability, risk, and resilience used consistently rather than interchangeably.
- A layered structure for locating where margin is consumed in your own systems, and therefore where intervention is effective.
- A clearer view of which environmental exposure is being retained rather than transferred, and why resilience investment is currently unpriced.
- A route to a comparable exposure baseline across a fleet, built from open environmental data, with no proprietary disclosure required.
- An open invitation to shape the OpRes framework before it becomes technical guidance.

Call to Action

- Problem to solve — no open method exists for relating environmental exposure to operational consequence across site, facility, IT, and fleet layers.
- Read and respond to the OpRes White Paper, Operational Resilience in Data Centers (v4.0, Draft for Consultation).
- Six priority questions are set out in Section 12.1.
- Get involved — OCP Sustainability Project, Operational Resilience workstream.
- Particularly seeking operators, thermal and power engineers, vendors, and insurance and risk-engineering practitioners.
- Responses to: maheshwari.neelam@ocproject.net
- OCP Sustainability Project: https://www.opencompute.org/community/sustainability

Open Discussion

